



MISO DPP 2023 Phase 1 Final Report

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Revision History

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Executive Summary

This report presents the results of a System Impact Study evaluating the interconnection of the generators in the Definitive Planning Phase (DPP) 2023 Phase 1. The study was conducted under MISO's direction and reviewed by an ad hoc study group. This group, composed of representatives from interconnection customers and transmission owners, was formed to assess the study's scope, methodology, models, and results.

- Phase 1 of the DPP 2023 analyzed 351 interconnection requests, totaling a nameplate capacity of 68,814.97 MW for Energy Resource Interconnection Service (ER or ERIS) and 62,711.62 MW of Network Resource Interconnection Service (NR or NRIS). Please refer to the *Executive Project and Upgrade Cost Summary* at the end of this report for an overview of non-CEII information such as Point of Interconnection, fuel type, service type, and total upgrade costs. Other non-CEII data may be obtained through the [MISO Public Queue](#). Full project details and modeling assumptions used in this analysis are provided in Appendix F (CEII) – Study Assumptions.

Screening analysis results for the DPP 2023 were made available via the MISO Extranet for Interconnection Customers on September 15, 2025. DPP 2023 kicked off for all regions on September 30, 2025. The draft DPP 2023 Phase 1 steady-state models were provided for ad-hoc review on September 30, 2025, with a deadline of October 14, 2025 for Interconnection Customers to complete the Appendix 10 of Attachment X Model Review form.

MISO utilized Pearl Street Technologies, Inc. SUGAR™ to:

- Build Bench and Study Models
- Update MTEP input files (.mon, .con, and .sub)
- Conduct ERIS thermal, ERIS voltage, and NRIS (Deliverability) Analysis
- Develop solutions and cost estimates in accordance with established [whitepapers](#) and [cost guides](#)
- Generate report appendices consistent with MISO practices.

The DPP 2023 Cycle Phase 1 study focused solely on steady-state analysis. Unless an NRIS Only or External NRIS¹ request was made, NRIS requests were also evaluated for ERIS. The results from the ERIS and NRIS analyses are recorded in Appendix C and Appendix D, respectively. Additionally, results for Transmission Owners with Local Planning Criteria (LPC) are documented in Appendix E (CEII) – LPC Results. The Phase 2 report will incorporate system impact studies from affected systems, as well as the MISO short circuit and stability analyses.

Significant upgrades are required to interconnect the DPP 2023 Phase 1 Interconnection Requests to the MISO transmission system. The full list of network upgrades required are detailed in Appendix B (CEII) – Network Upgrade Summary. The cost responsibility for these upgrades have been allocated in accordance with MISO BPM-015. Please refer to Appendix A (CEII) for the breakdown cost for each Interconnection Request and the associated milestone payments required to advance to the next phase.

Steady State Thermal Analyses

Steady-state ERIS and NRIS (deliverability) analyses are conducted to identify what network upgrades, if any, are required to reliably interconnect current study interconnection requests to the MISO transmission system. Section 6 of the MISO BPM-015 details the processes and methodologies used for the DPP steady-state thermal analysis. Please note that for Phase 1 network upgrade development was limited to the criteria as laid out in the [Transmission Mitigation Selection and Cost Estimation Approach Whitepaper](#).

ERIS Analysis

The following models were developed for the ERIS analysis:

¹ HVDC projects with External NRIS are studied for ERIS in accordance with Attachment GGG of the MISO Tariff.



- Summer Shoulder Bench
- Summer Shoulder Discharging Study
- Summer Shoulder Charging Study
- Summer Peak Bench
- Summer Peak Discharging

• The DPP 2023 Cycle 1 models originated from the MTEP24 models, with the Bench Cases including all higher-queued interconnection requests. Please refer to Appendix F (CEII) – Study Assumptions *Higher Queued Network Upgrades* for the list of higher queued network upgrades included in the analysis. Study Cases contain all the interconnection requests in DPP 2023 Phase 1 in addition to all the facilities in the Bench Cases.

MISO utilized PSS®E 35.6, Pearl Street Technologies, Inc. SUGAR™, and PowerGEM TARA v2202.2.0.0 for steady-state analysis. AC Contingency calculations were performed on the bench and study models and identified network constraints in accordance with the MISO BPM-015. The full list of constraints identified in the ERIS and NRIS analysis are detailed in Appendix C (CEII) – ERIS Results and Appendix D (CEII) – Deliverability Results.

Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact – no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2, P4, P5, and P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

Network upgrades and associated costs to mitigate constraints observed in the system impact study were developed by the SUGAR software according to established mitigation guidelines. Transmission owners may not have comprehensively reviewed all proposed network upgrades. Network upgrade costs were allocated in accordance with the MISO BPM-015. The list of network upgrades required for interconnection service are detailed in Appendix B (CEII) – Network Upgrade Summary. Cost allocation of those upgrades along with the required milestone payments to advance to the next phase are outlined in Appendix A (CEII) – Cost Allocation and Milestone Payment Summaries.

Network upgrade cost estimates were developed on a planning level and are later refined in Phase 2 facility studies. Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator. Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.

MISO will evaluate interconnection requests for FERC 827 requirements in Phases 2 and 3.

JTIQ Expanded Scope Analysis

An expanded scope analysis was conducted in accordance with the MISO-SPP JOA to determine interconnection requests which have impacts on the Affected System greater than the specified criteria to



enter into an appropriate agreement with the Affected System to address such impacts in accordance with the rules of the Affected System.

NRIS (Deliverability) Analysis

Generator interconnection projects must pass the Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. The Generator Deliverability Study ensures that Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak conditions without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

The model developed for the NRIS analysis was derived from the Summer Peak Discharging case from the ERIIS analysis; however, the study units are not dispatched as this occurs through the TARA deliverability analysis.

Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a distribution factor (DF) greater than 5%, part or all its output is not deliverable. The restricted MW is calculated as follows:

$$(MW\ restricted) = \frac{worst\ loading - MW\ rating}{(generator\ sensitivity\ factor)}$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

Shared Network Upgrades Analysis

The Shared Network Upgrade (SNU) Analysis, which tests Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study. Please refer to Appendix A (CEII) – Cost Allocation and Milestone Payment Summaries for maximum MW impacts and SNU cost allocations. Details regarding the analysis are provided in Appendix H (CEII) – Shared Network Upgrade (SNU) Results.

Evaluation of Alternative Network Upgrades

MISO provided the ad-hoc group 15 Business Days to supply alternative network upgrades for evaluation in the system impact study. If the alternative mitigation was provided in accordance with the instructions provided, the proposal was reviewed by MISO and Pearl Street. Alternative mitigations were rejected if the proposal:

- failed to provide sufficient evidence that the mitigation would alleviate the observed constraint,
- required further coordination with affected Transmission Owners to determine viability of the mitigation,
- supplied conflicting, duplicate, or erroneous data.

Alternative mitigations which passed visual inspection were provided as an input to SUGAR™. If the constraint was observed in the analysis, SUGAR™ used the alternative mitigation and cost estimate in the report. If the constraint was not observed or the mitigation or was not compatible with the tool, it was not included in the analysis.

Local Planning Criteria (LPC)

Ameren



The Ameren LPC is evaluated in stability only.

ATC

The ATC LPC study is performed to adhere to ATC Local Planning Criteria. Details regarding this LPC are available in Appendix E (CEII) – LPC Results.

Dairyland Power Cooperative

DPC elected not to perform an LPC analysis for phase 1. An LPC analysis will be performed for Phase 2 and 3.

Entergy

The Entergy LPC is evaluated in stability only.

Great River Energy

GRE elected not to perform an LPC analysis for phase 1. An LPC analysis will be performed for Phase 2 and 3.

ITC

The ITC LPC is evaluated in stability only.

MDU

MDU elected not to perform an LPC analysis for phase 1. An LPC analysis will be performed for Phase 2 and 3.

NIPSCO

LPC results for NIPSCO are available in Appendix E (CEII) – LPC Results.

OTP

OTP elected not to perform an LPC analysis for phase 1. An LPC analysis will be performed for Phase 2 and 3.



Appendices

Appendix A (CEII) – Cost Allocation and Milestone Payment Summaries

- Milestone Payment Summary
- Backbone Upgrade Cost Allocation Summary
- ERIS Thermal Cost Allocation Summary
- ERIS Voltage Cost Allocation Summary
- JTIQ Expanded Scope Cost Allocation Summary
- LPC Cost Allocation Summary
- NRIS Cost Allocation Summary
- Interconnection Facilities Cost Summary
- Shared Network Upgrade Cost Allocation Summary

Appendix B (CEII) – Network Upgrade Summary

Appendix C (CEII) – ERIS Results

- ERIS Thermal Analysis
- ERIS Thermal Cost Allocation
- ERIS Voltage Analysis
- ERIS Voltage Cost Allocation

Appendix D (CEII) – Deliverability Results

Appendix E (CEII) – LPC Results

- ATC
- NIPSCO

Appendix F (CEII) – Study Assumptions

- Generator Dispatch
- Higher Queued Network Upgrades
- Interconnection Facility Cost Estimates
- LRTP Projects
- Point of Interconnection (POI) Modeling
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Appendix G (CEII) – JTIQ Expanded Scope Results

Appendix H (CEII) – Shared Network Upgrade (SNU) Results